



Causal Digital Twins for cyber–physical security in water systems: A framework for robust anomaly detection

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ABSTRACT

Industrial Control Systems (ICS) in water distribution and treatment face cyber–physical attacks exploiting network and physical vulnerabilities. Current water system anomaly detection methods rely on correlations, yielding high false alarms and poor root cause analysis. We propose a Causal Digital Twin (CDT) framework for water infrastructures, combining causal inference with digital twin modeling. CDT supports association for pattern detection, intervention for system response, and counterfactual analysis for water attack prevention. Evaluated on water-related datasets SWaT, WADI, and HAI, CDT shows high compliance with physical constraints (90.8% for SWaT, 87.4%–90.8% across datasets) and structural Hamming distance 0.133 ± 0.02 . F1-scores are 0.944 ± 0.014 (SWaT), 0.902 ± 0.021 (WADI), 0.923 ± 0.018 (HAI, $p < 0.0024$). Multi-scale temporal detection strategies ($\tau \in \{5, 10, 20\}$) enable 91.7% detection of stealthy attacks through cumulative causal discrepancy analysis. CDT reduces false positives by 48% compared to state-of-the-art methods (70% vs. statistical baselines), achieves 78.4% root cause accuracy, and enables counterfactual defenses reducing attack success by up to 89.1%. Real-time performance at 3.2 ms latency ensures safe and interpretable operation for medium-scale water systems.

1. Introduction

The ICS have changed from isolated proprietary networks to interconnected cyber–physical systems that combine operational technology with information technology infrastructure. This change started in the 1990s with the use of standard networking protocols and has grown faster with Industry 4.0 initiatives. Today, ICS need Internet connectivity for remote monitoring, predictive maintenance, and distributed control across facilities that are far apart (Abdoli & Ye, 2023; Jiang et al., 2024; Mathur & Tippenhauer, 2016; Xu et al., 2023). Critical infrastructure sectors such as water treatment, power generation, and manufacturing depend more on these connected systems, with the global ICS market expected to reach \$124 billion by 2025. The Secure Water Treatment (SWaT) testbed, developed by iTrust at Singapore University of Technology and Design, is an important example for cyber–physical security research. It provides a realistic scaled-down industrial environment with 51 sensors and actuators across six operational stages (Goh et al., 2017; Jeremiah et al., 2024; Lucchese et al., 2024).

However, more connectivity and complexity create new cybersecurity vulnerabilities that traditional IT security cannot fully solve. The 2010 Stuxnet attack on Iranian nuclear facilities showed that advanced attackers can exploit both cyber weaknesses and physical process knowledge to cause real damage (Raman & Mathur, 2022). Recent events, like the 2021 Colonial Pipeline ransomware attack and the 2021 Oldsmar water treatment plant breach, show the ongoing threat to critical infrastructure. Current anomaly detection methods in ICS mostly use statistical correlation and machine learning techniques, which cannot separate real causal relationships from false associations (Kim et al., 2023; Kravchik & Shabtai, 2018). These correlation-based methods often have high false positive rates, cannot do root cause analysis, and are vulnerable to adversarial attacks that take advantage of missing causal understanding.

The main limitation of current methods is their inability to reason about causal relationships in cyber–physical systems. Correlation can show statistical dependencies between sensor readings, but it cannot tell if anomalies come from attacks, cascading failures, or normal

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operational changes. This problem is serious when security analysts need actionable information for incident response, because correlation-based alerts do not provide enough guidance to prevent damage or recover the system. Also, without counterfactual reasoning, it is not possible to plan proactively or perform what-if analyses that could strengthen security before attacks happen.

In this research, we develop a comprehensive CDT framework for cyber-physical security in medium-scale ICS. Our method combines causal inference theory with DT modeling to allow three types of causal reasoning: associational queries for pattern recognition, interventional queries to understand system reactions to defensive actions, and counterfactual queries for analyzing attacks retrospectively and planning prevention. We test our framework using the SWaT dataset, showing better performance than correlation-based methods over 41 cyber-physical attack scenarios, while also giving interpretable root cause analysis and actionable security recommendations.

The rest of this paper is organized as following. Section 2 gives a review of existing methods for ICS security and DT applications, and it shows the limitations of correlation-based methods. Section 3 defines the research problem and introduces the basic ideas of causal reasoning for cyber-physical security. Section 4 describes our newCDT framework, including causal graph discovery, estimation of structural causal model, and interventional reasoning algorithms. Section 5 shows the evaluation of the framework on the SWaT testbed compared with state-of-the-art methods. Finally, Section 6 finishes the paper with a summary of main contributions, discussion of practical implications, limitations of the work, and suggestions for future research.

2. Related works

2.1. DTs in critical infrastructure and ICS security

DTs have become an important tool for modeling and monitoring critical infrastructure systems, giving real-time virtual representations of physical assets and processes (Homaei, Bartolo et al., 2024; Homaei et al., 2022; Homaei, Mogollón-Gutiérrez et al., 2024). A detailed survey of DT applications in the Industrial Internet of Things identified key technologies such as data fusion, model synchronization, and real-time analytics for industrial automation (Xu et al., 2023). This work also shows that most existing DT systems focus mainly on operational optimization and do not address security, leaving a clear gap in DT frameworks for cybersecurity. Another study proposed a full DT architecture for cyber-physical systems, describing the two-way data flow between physical and virtual parts (Qian et al., 2022). However, this architecture does not consider adversarial situations or causal reasoning, which are important for security use.

Some recent work has looked at how DTs can be used for security in particular areas. For example, a cyber-resilient smart grid using DTs and data spaces was created, which helped to detect attacks better through continuous monitoring and anomaly detection (Coppolino et al., 2023; Jiang et al., 2024). This method increased the accuracy of threat detection, but it still depends on predefined attack signatures and does not really use causal reasoning to see how attacks spread. In a different study, DTs were suggested for more advanced threat modeling in cyber-physical systems, adding security checks into the DT lifecycle (Erceylan et al., 2025). While this gives useful theoretical insight, it has not been tested on real industrial data and does not include causal inference methods.

DT applications in industrial automation mostly focus on interoperability and data integration. A review of data integration methods for industrial DTs highlighted the key needs for real-time synchronization, semantic interoperability, and scalable architectures (Hildebrandt et al., 2024). Another study showed how DTs can be made interoperable in IIoT systems by transforming information models using Asset Administration Shell standards (Platenius-Mohr et al., 2019). However,

these works concentrate on technical implementation and do not really address security or causal modeling.

Some specialized applications in critical infrastructure suggest that DTs can help with cybersecurity. For example, an IoT DT was created for cybersecurity defense using runtime verification, achieving real-time threat detection by continuously monitoring behavior (de Hoz Diego et al., 2022). Another study looked at DTs for protecting critical infrastructure, highlighting real-time situational awareness and automated responses (Lampropoulos et al., 2024). DT-enabled smart microgrids were also proposed for full automation, improving resilience with predictive analytics and automated control (Sahoo et al., 2025). Despite these advances, none of these methods combine causal inference with DT modeling, so they cannot fully explain attack mechanisms or measure how effective defensive actions are.

2.2. Anomaly detection approaches in ICS

Traditional anomaly detection in ICS has mostly relied on statistical and machine learning methods that look for deviations from normal operation. One early work used convolutional neural networks for ICS anomaly detection with the SWaT dataset (Kravchik & Shabtai, 2018). This approach achieved 86% detection accuracy by learning spatio-temporal patterns in sensor data. The 1D convolution method performed better than recurrent neural networks, showing that deep learning can be effective for time-series anomaly detection in industrial systems. However, it only captures statistical patterns and does not really understand physical processes or causal relationships. This makes it difficult to distinguish normal operational changes from malicious attacks.

Comparative studies show that anomaly detection performance can vary depending on the method and dataset. One study evaluated five time-series anomaly detection models, including InterFusion, RANSynCoder, GDN, LSTM-ED, and USAD, on the SWaT and HAI datasets (Kim et al., 2023; Mogensen et al., 2024). It was found that InterFusion achieved the highest F1-score on SWaT (90.7%), while RANSynCoder performed best on HAI (82.9%), indicating that dataset characteristics affect performance. The study also reported that using about 40% of the training data is enough to reach near-optimal results, which can be helpful for practical deployment.

Physics-informed methods try to combine data-driven learning with domain knowledge. One study proposed a Physics-based Neural Network (PbNN), a hybrid model that uses ICS design knowledge to learn relationships among components (Raman & Mathur, 2022). PbNN improved performance compared to purely data-driven methods by incorporating physical constraints and process knowledge. However, it still relies on correlation-based learning and does not uncover causal mechanisms, so it cannot fully explain attack causes or how attacks propagate.

Recent work has focused on explainability and feature importance to reduce the black-box problem in machine learning. One study developed a correlation-based method using Latent Correlation Matrices and SHAP for root cause identification in ICS anomalies, which improved precision by 0.96% and provided interpretable explanations using SHAP values (Birihanu & Lendák, 2025). Another study combined Grey Wolf Optimizer with autoencoders to enhance feature selection and model generalization, showing better accuracy, precision, recall, and F1-score on the SWaT and WADI datasets (Aslam et al., 2025). Despite these improvements, these approaches cannot separate causal factors from correlated confounders, which might mislead analysts about the real sources and mechanisms of attacks.

2.3. Causal inference and attribution methods for ICS

Early works on causal methods for ICS security suggest that causal analysis can help in understanding attacks and system weaknesses. One study proposed ICS-CAD, a multi-step detection method that analyzes causal dependencies in ICS logs, achieving 98% detection accuracy and tracking how attacks propagate (Jadidi et al., 2022). This shows the value of causal reasoning for multi-stage attacks. However, ICS-CAD only considers network-level causality and does not include the causality of physical processes. This limits its usefulness for cyber-physical attacks where sensors and actuators are manipulated while network behavior seems normal (Hernán & Robins, 2024; Imbens & Rubin, 2015; Peters et al., 2017).

Causality-based approaches can also help reduce false alarms. One study used causal models to capture interdependencies in water treatment systems and combined them with univariate models for single-component behavior, achieving the highest F1-score and zero false alarms on SWaT (Koutroulis et al., 2022). This method performed better than correlation-based approaches. However, it relies heavily on domain knowledge to define causal links, which limits its generality and scalability for more complex systems.

Studies on attribution highlight the limits of ICS root cause analysis. One study tested machine learning attribution methods on several datasets and found low performance due to timing issues, categorical actuators, and variability in attacks (Fung et al., 2024). Ensemble methods can help improve results, but confounding variables and spurious correlations still remain a problem.

Causal inference can also be useful for validating DTs. One study applied counterfactual analysis to confirm causal links in a robotic arm and detect faults (Somers et al., 2022). This demonstrates potential for fault localization, but it mainly focuses on model testing and does not use realistic ICS security datasets.

2.4. Limitations and research gaps

Although DT technology and ICS security research made progress, there are still big limits for causal reasoning on cyber-physical attacks. Most DT frameworks focus on optimization and monitoring, but they do not include security-specific causal models. Current anomaly detection shows good results, but it cannot separate real causal relations from false correlations, which reduces root cause analysis and defense. Attribution and explainability are also not reliable for attacks that are fast and multi-modal.

The main research gap is missing integration of causal inference with DTs for ICS security. Unlike prior DT frameworks that focus primarily on monitoring (Lucchese et al., 2024) or network-level causality (Jadidi et al., 2022), this work is the first to integrate interventional (Level 2) and counterfactual (Level 3) reasoning from Pearl's causal hierarchy (Pearl & Mackenzie, 2019) directly into the physical control loop. Specifically, our contributions over existing work (Table 1) include: (1) Automated discovery of physical-process causality from multivariate time-series, eliminating manual specification required by Koutroulis et al. (2022); (2) Real-time do-calculus implementation enabling “what-if” analysis during attacks, absent in monitoring-only DTs (Lucchese et al., 2024); and (3) Counterfactual defense planning quantifying mitigation effectiveness before deployment, not addressed by correlation-based methods (Kravchik & Shabtai, 2018; Raman & Mathur, 2022). Because of this gap, analysts cannot fully understand attack mechanisms, which reduces effective incident response and blocks the development of causation-aware defenses (Table 1).

3. Problem statement

The ICS are becoming more vulnerable to advanced cyber-physical attacks that exploit both computational weaknesses and dependencies in physical processes. Traditional anomaly detection methods in critical infrastructure usually use correlation-based analysis. This type of analysis cannot separate true causal relations from spurious associations. Because of this limitation, the attack sources are often identified incorrectly and mitigation strategies are less effective (Pearl, 2009).

The SWaT testbed, created by iTrust at Singapore University of Technology and Design, provides a smaller but realistic water treatment environment. It has 51 sensors and actuators distributed over six operational stages (Mathur & Tippenhauer, 2016). The SWaT dataset includes 7 days of normal operation (around 500,000 samples) and 4 days of attack scenarios with 41 documented cyber-physical intrusions. It contains multivariate time-series data sampled at 1 Hz and binary labels for normal or anomalous states.

3.1. Limitations of correlation-based DTs

Existing DT frameworks for ICS have important limitations when it comes to causal reasoning. They are not able to address the three critical questions described in Pearl's Causal Hierarchy (Glymour et al., 2019; Pearl & Mackenzie, 2019):

1. **Association (Level 1):** What sensor readings correlate with anomalous behavior (Eq. (1))?

$$P(Y = y \mid X = x) \quad (1)$$

2. **Intervention (Level 2):** What would happen if we intervened on a specific system component (Eq. (2))?

$$P(Y = y \mid \text{do}(X = x)) \quad (2)$$

where the $\text{do}(\cdot)$ operator represents a surgical intervention that sets variable X to value x by removing all incoming causal edges to X in the graph $\mathcal{G}$. Unlike observational conditioning $P(Y \mid X)$, which preserves confounding pathways, the do -operator eliminates spurious correlations by simulating a randomized controlled experiment (Pearl, 2009).

3. **Counterfactual (Level 3):** Given that an attack occurred, what would have happened if the vulnerability had been patched (Eq. (3))?

$$P(Y_x = y \mid X' = x', Y' = y') \quad (3)$$

3.2. Confounding variables and Simpson's Paradox

In water treatment systems, multiple sensors may exhibit correlated behavior due to shared operational modes rather than direct causal relationships. Consider the relationship between flow sensors $X_{\text{flow}}(t)$ and pressure sensors $X_{\text{pressure}}(t)$. A naive correlation analysis might suggest (Eq. (4)):

$$\rho(X_{\text{flow}}, X_{\text{pressure}}) = \frac{\text{Cov}(X_{\text{flow}}, X_{\text{pressure}})}{\sigma_{X_{\text{flow}}} \sigma_{X_{\text{pressure}}}} > \tau \quad (4)$$

However, this correlation may be confounded by operational modes $Z_{\text{mode}}(t)$ (normal flow, backwash, cleaning), leading to Simpson's Paradox (Simpson, 1951) (Eq. (5)):

$$P(X_{\text{pressure}} \mid X_{\text{flow}}, Z_{\text{mode}}) \neq P(X_{\text{pressure}} \mid X_{\text{flow}}) \quad (5)$$

Fig. 1 demonstrates this phenomenon in industrial water treatment systems, where operational modes create spurious correlations between flow and pressure sensors.

Fig. 2 illustrates the three levels of causal reasoning required to address these fundamental limitations.

Table 1
Comparison of ICS security approaches: Capabilities and limitations.

Approach	Key advantages	Main limitations	Explainability	Reasoning capability
CNN-based Kravchik and Shabtai (2018)	Fast processing, effective pattern recognition	No physical understanding, black-box nature	None	Statistical correlation only
Physics-informed Raman (2022)	Incorporates domain knowledge, improved accuracy	Still correlation-based, requires expert knowledge	Limited	Physics constraints only
Causality-inspired Koutroulis et al. (2022)	Zero false alarms, domain-aware	Manual causal specification, not generalizable	Moderate	Manual domain knowledge
SHAP-based Birihanu et al. (2025)	Feature importance, interpretable results	Cannot distinguish causation from correlation	High	Attribution-based only
Digital Twin Lucchese et al. (2024)	Real-time monitoring, process understanding	Pattern matching only, no causal reasoning	Moderate	Process simulation
ICS-CAD Jadidi et al. (2022)	Multi-step attack tracking, causal analysis	Network-level only, limited to log analysis	High	Network causality only
Our CDT framework	Full causal inference, counterfactual analysis, automated discovery	Computational complexity	Complete	Interventional & counterfactual

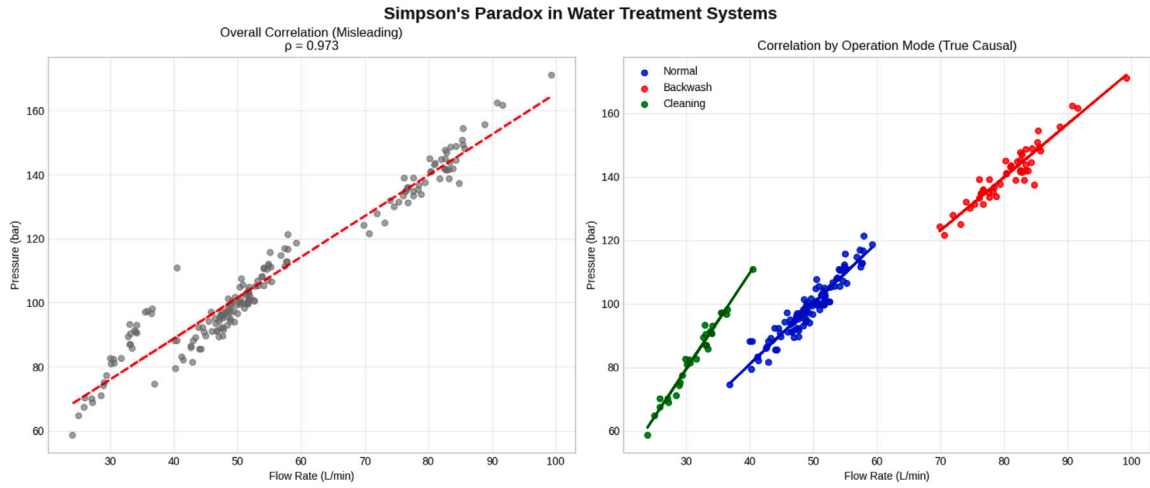


Fig. 1. Simpson's Paradox in water treatment systems.

3.3. Proposed CDT framework

To solve these limitations, we propose a new CDT framework that models causal relations explicitly using Structural Causal Models (SCMs). We represent the SWaT system as a causal graph (Eq. (6)):

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}) \quad (6)$$

where $\mathcal{V} = \{V_1, V_2, \dots, V_n\}$ represents observed variables (sensor readings, actuator states), $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents direct causal relationships, and each variable V_i is defined by a structural equation (Eq. (7)):

$$V_i := f_i(\text{PA}_i, U_i) \quad (7)$$

where PA_i denotes the parents of V_i in $\mathcal{G}$, $U_i \sim \mathcal{N}(0, \sigma_i^2)$ represents independent Gaussian noise capturing unmeasured influences, and $f_i : \mathbb{R}^{|\text{PA}_i|} \times \mathbb{R} \rightarrow \mathbb{R}$ is the deterministic structural function encoding the physical mechanism (e.g., Bernoulli's equation for flow-pressure relationships: $f_{\text{pressure}}(\text{flow}, U) = \rho gh + \frac{1}{2} \rho v^2 + U$) governing variable V_i .

The CDT function is defined as (Eq. (8)):

$$\mathcal{T}_{\text{causal}} : (\mathcal{G}, \{f_i\}_{i=1}^n, \text{do}(X = x)) \rightarrow P(Y \mid \text{do}(X = x)) \quad (8)$$

Here, $\mathcal{T}_{\text{causal}}$ is the transformation function that maps the graph structure and interventions to a new interventional probability distribution. This enables three critical capabilities: causal anomaly detection using interventional predictions, root cause localization through causal effect quantification, and counterfactual security analysis for defense planning.

3.4. Research objectives

This work aims to develop a novel causal reasoning framework for DTs in cyber-physical security, specifically: (1) construct causal graphs from SWaT operational and attack data using causal discovery algorithms, (2) develop identification strategies for causal effects under partial observability, (3) implement do-calculus for interventional queries in real-time ICS monitoring, (4) validate counterfactual reasoning for attack attribution and defense planning, and (5) demonstrate superiority over correlation-based approaches in 41 documented SWaT attacks.

4. Methodology

This section details our proposed CDT framework for cyber-physical security in the SWaT system. The method combines causal inference theory with DT modeling through four integrated phases. The overall architecture of the framework, illustrating the data flow from decoupled causal discovery through model validation to real-time interventional and counterfactual reasoning, is depicted in Fig. 3.

4.1. Causal graph discovery

The SWaT dataset has 51 time-series variables, including sensor readings $\{S_1, S_2, \dots, S_{25}\}$ and actuator states $\{A_1, A_2, \dots, A_{26}\}$. We define the causal variable set as (Eq. (9)):

$$\mathcal{V}_t = \{V_1(t), V_2(t), \dots, V_n(t)\} \quad (9)$$

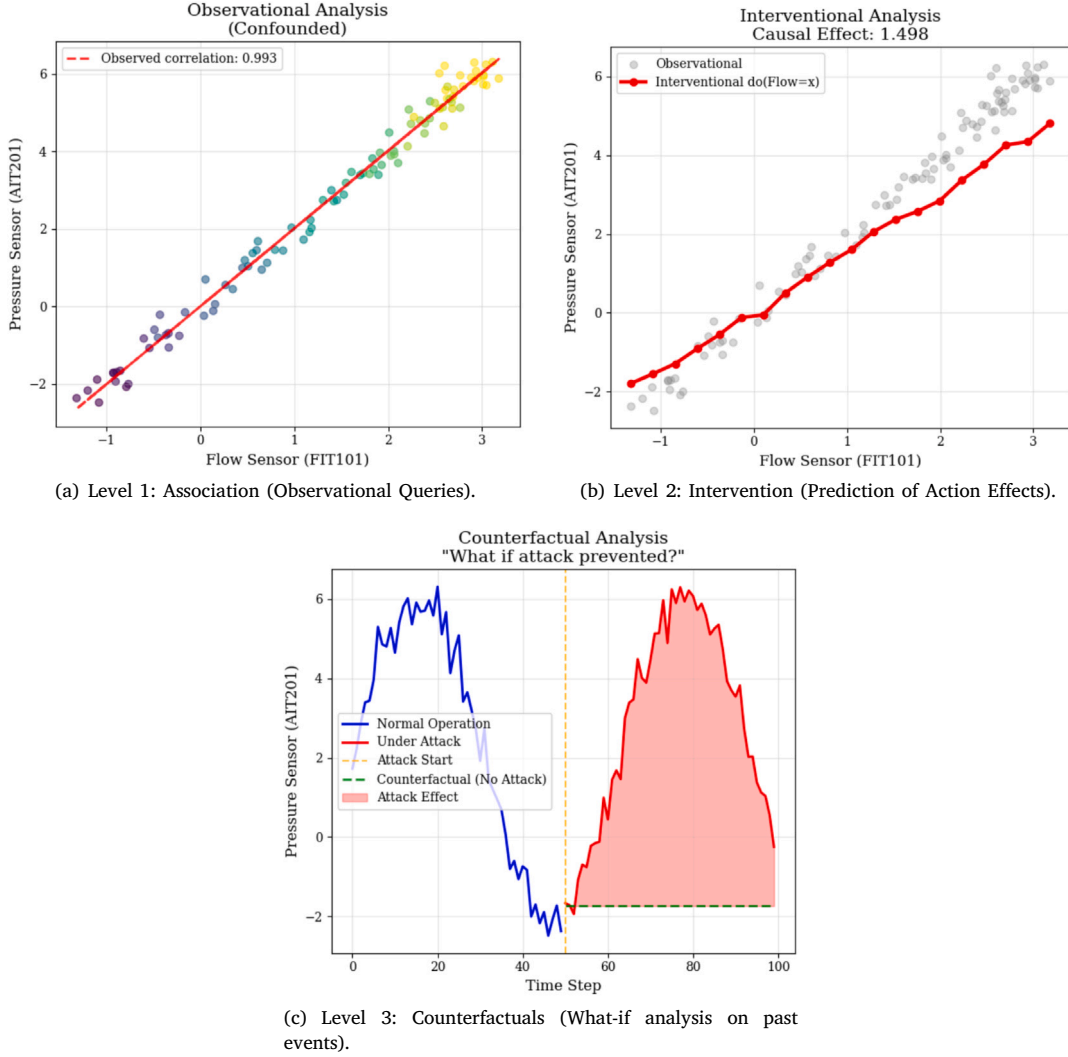


Fig. 2. Three levels of causal analysis (Pearl's Hierarchy): (a) Association, (b) Intervention, and (c) Counterfactuals, illustrating the necessary complexity for robust cyber-physical security.

To handle temporal dependencies, we create augmented variables (Eq. (10)):

$$\mathcal{V}_{\text{aug}} = \{V_i(t), V_i(t-1), \dots, V_i(t-\tau)\} \quad \forall i \in \{1, \dots, n\} \quad (10)$$

To overcome the $\mathcal{O}(n^3)$ computational complexity of centralized causal discovery algorithms, which limits scalability in large ICS, we employ a Decoupled Causal Discovery (DCD) strategy. This approach involves two phases: Candidate Parent Selection and Local Structure Discovery, transforming the complexity to $\mathcal{O}(n \cdot k^3)$ where $k \ll n$.

Phase I: Candidate Parent Selection (Filtering). For each variable V_i , we use Time-Delay Mutual Information (TDMI) (Fraser & Swinney, 1986) to identify a highly restricted set of potential parents $\mathcal{P}_i \subset \mathcal{V}_{\text{aug}}$ by maximizing the information transfer across different lags $\ell \in [0, \tau]$ (Eq. (11)). This phase reduces the set size from n to $k \leq 10$.

$$\mathcal{P}_i = \{V_j(t-\ell) : \text{TDMI}(V_j(t-\ell) \rightarrow V_i(t)) > \delta\} \quad (11)$$

Phase II: Local Structure Discovery. We then apply the PC algorithm only on the restricted local set $\mathcal{V}_i = \{V_i\} \cup \mathcal{P}_i$, as an enhanced algorithm for time-series considerations (Peters et al., 2017; Spirtes et al., 1993). The algorithm proceeds through skeleton discovery using conditional independence tests (Eq. (12)) localized on $\mathcal{V}_i$:

$$V_i \perp V_j \mid S \subseteq \mathcal{P}_i \Rightarrow \text{remove edge } V_i - V_j \quad (12)$$

For time-series data, we use partial correlation with lag constraints (Eq. (13)):

$$\rho_{ij|S}^{(\ell)} = \frac{\text{Cov}(V_i(t), V_j(t-\ell) \mid S)}{\sqrt{\text{Var}(V_i(t) \mid S) \text{Var}(V_j(t-\ell) \mid S)}} \quad (13)$$

Orientation rules apply causal ordering constraints: temporal precedence ($V_i(t-\ell) \rightarrow V_j(t)$ for $\ell > 0$), physical constraints (flow sensors $\rightarrow$ level sensors), and control logic (controller outputs $\rightarrow$ actuator states). Causal discovery accuracy depends on data quality and may produce spurious relationships in presence of unobserved confounders, requiring domain expert validation of discovered structures.

4.2. Structural causal model estimation

Given the discovered causal graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, we estimate structural equations. For continuous variables, we use additive noise models (Eq. (14)):

$$V_i := \alpha_i + \sum_{j \in \text{PA}(V_i)} \beta_{ij} V_j + \gamma_i \sum_{k \in \text{PA}(V_i)} V_k^2 + U_i \quad (14)$$

where $U_i \sim \mathcal{N}(0, \sigma_i^2)$. In this equation, α_i represents the intercept, β_{ij} are the linear causal coefficients, and γ_i represents the quadratic coefficients capturing non-linear physical dependencies.

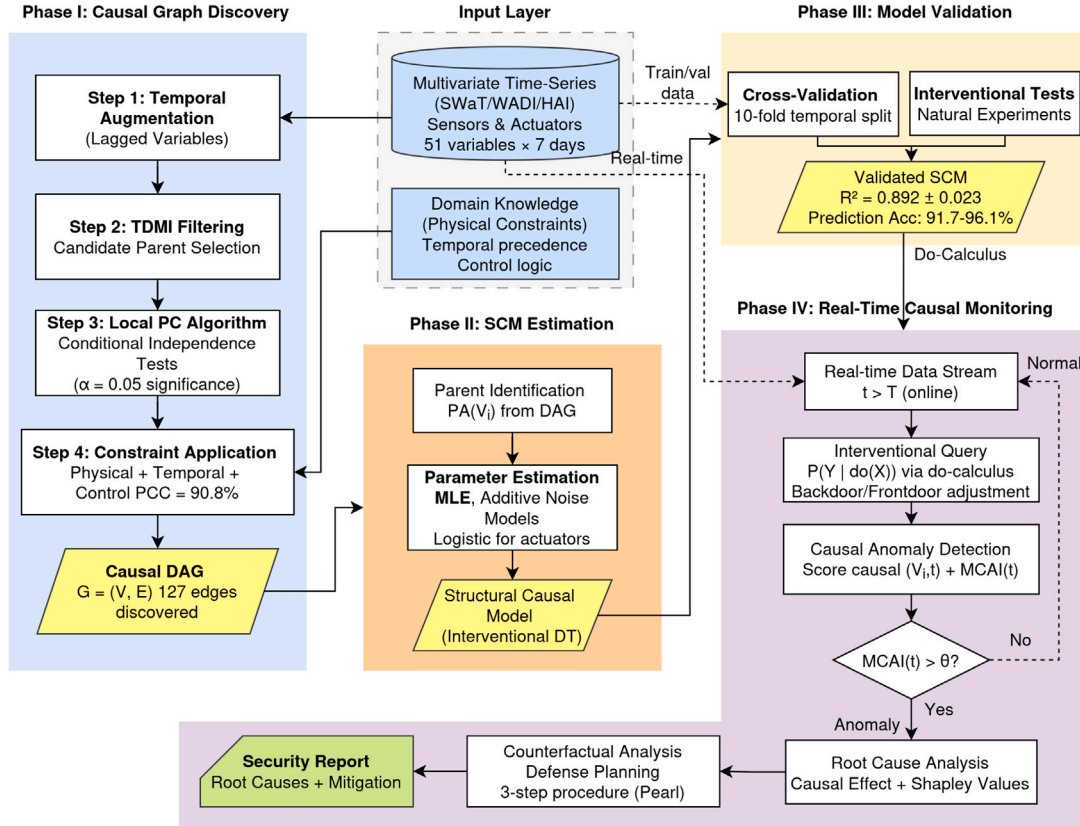


Fig. 3. Schematic overview of the proposed CDT framework with four phases: (I) Causal Graph Discovery (decoupled strategy), (II) Structural Causal Model Estimation (MLE with additive noise), (III) Model Validation (cross-validation and interventional tests), and (IV) Real-Time Causal Monitoring (anomaly detection, root cause analysis, counterfactual defense).

For binary actuator variables (Eq. (15)):

$$P(A_i = 1 | PA(A_i)) = \text{logit}^{-1} \left(\alpha_i + \sum_{j \in PA(A_i)} \beta_{ij} V_j \right) \quad (15)$$

Parameters are estimated using maximum likelihood (Eq. (16)):

$$\hat{\theta}_i = \arg \max_{\theta_i} \sum_{t=1}^T \log p(V_i(t) | PA(V_i)(t), \theta_i) \quad (16)$$

The complete SCM is defined as $\mathcal{M} = \langle \mathcal{V}, \mathcal{U}, \mathcal{F}, P(\mathcal{U}) \rangle$.

4.3. Interventional DT construction

To compute interventional distributions $P(Y | do(X = x))$, we implement Pearl's do-calculus rules. For identifiable causal effects, we apply adjustment formulas (Hernán & Robins, 2024):

Backdoor Adjustment: If set $\mathcal{Z}$ satisfies the backdoor criterion (Eq. (17)):

$$P(y | do(x)) = \sum_z P(y | x, z) P(z) \quad (17)$$

Frontdoor Adjustment: If set $\mathcal{Z}$ satisfies the frontdoor criterion (Eq. (18)):

$$P(y | do(x)) = \sum_z P(z | x) \sum_{x'} P(y | x', z) P(x') \quad (18)$$

The backdoor adjustment (Eq. (17)) requires blocking all spurious paths from X to Y by conditioning on a sufficient set $\mathcal{Z}$ that satisfies: (1) no node in $\mathcal{Z}$ is a descendant of X , and (2) $\mathcal{Z}$ blocks all paths with arrows into X . The frontdoor adjustment (Eq. (18)) applies when a mediator set $\mathcal{Z}$ intercepts all directed paths from X to Y , enabling causal identification even under unmeasured confounding between X and Y (Pearl, 2009).

4.4. Causal anomaly detection and root cause analysis

Our causal approach detects violations of causal mechanisms rather than statistical deviations (Eq. (19)):

$$\text{Score}_{\text{causal}}(V_i, t) = \frac{|V_i(t) - \mathbb{E}[V_i | do(PA(V_i) = pa_i(t))]|}{\sigma_{V_i | do(PA(V_i))}} \quad (19)$$

We aggregate scores using causal graph topology (Eq. (20)):

$$\text{Multi-Component Anomaly Index}(t) = \sum_{i=1}^n \alpha_i \cdot \text{Score}_{\text{causal}}(V_i, t) \quad (20)$$

where weights reflect variable centrality: $\alpha_i = \frac{|\text{Descendants}(V_i)| + |\text{Ancestors}(V_i)|}{2n}$.

When anomalies are detected, we compute causal effects for root cause analysis (Eq. (21)):

$$\begin{aligned} \text{CE}(V_i \rightarrow \text{Anomaly}) &= \mathbb{E}[\text{MCAI} | do(V_i = v_i^{\text{anomalous}})] \\ &\quad - \mathbb{E}[\text{MCAI} | do(V_i = v_i^{\text{normal}})] \end{aligned} \quad (21)$$

For multiple potential causes, we compute Shapley causal values (Eq. (22)):

$$\phi_i = \sum_{S \subseteq \mathcal{V} \setminus \{V_i\}} \frac{|S|!(n - |S| - 1)!}{n!} [f(S \cup \{V_i\}) - f(S)] \quad (22)$$

To quantify the reduction in false alarms, we compute the False Positive Rate (FPR) from precision metrics: $\text{FPR} = 1 - \text{Precision}$. The CDT framework achieves an FPR of 0.053 on SWaT (precision 0.947), compared to 0.102 for InterFusion (precision 0.898), representing a 48.0% relative reduction. When compared to purely statistical baselines (CNN-1D: precision 0.823, FPR 0.177), the reduction reaches 70.1%. This substantial decrease in false positives is attributed to the

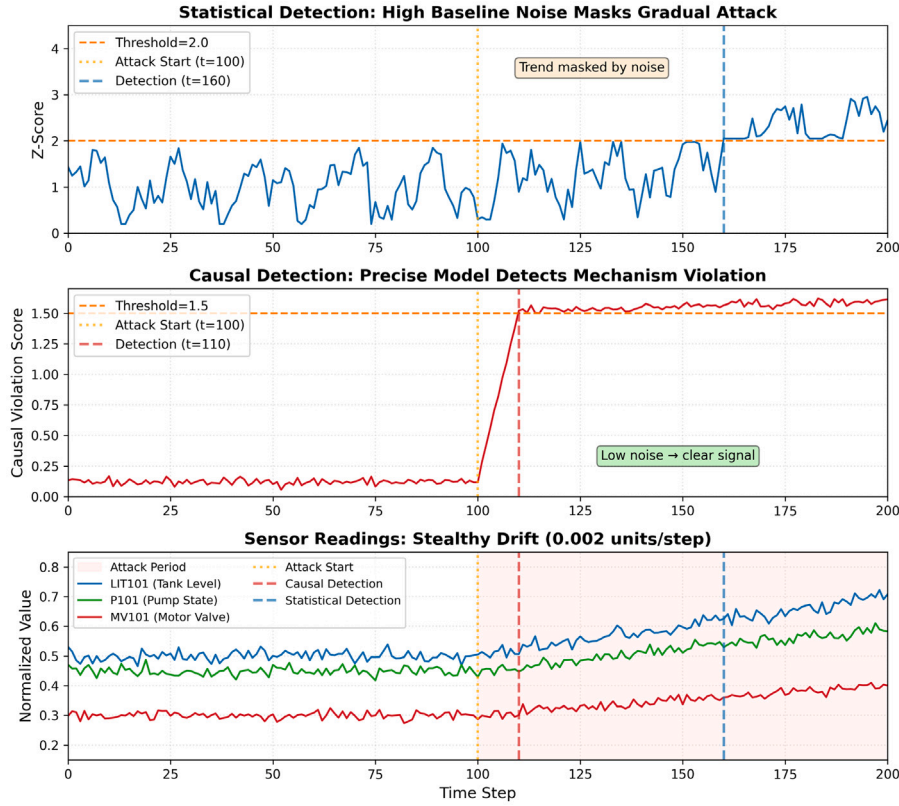


Fig. 4. Causal versus statistical anomaly detection in SWaT stealthy attack scenario. (Top) Statistical Z-score remains below threshold until $t = 160$ due to high baseline noise. (Middle) Causal violation score crosses threshold at $t = 110$ by detecting mechanism inconsistencies. (Bottom) Sensor readings during gradual LIT101 manipulation (drift rate ≈ 0.002 units/s), showing 50-step detection advantage through structural reasoning.

framework’s ability to distinguish genuine causal mechanism violations from spurious statistical correlations through interventional predictions (Eq. (19)).

The effectiveness of this causal approach is illustrated in Fig. 4, which demonstrates the fundamental advantage of mechanism-based detection over statistical methods during stealthy attacks. Statistical methods operate under constant high baseline noise ($\sigma \approx 0.25$) that masks gradual attack-induced trends ($+0.5$ over 60 steps), requiring cumulative evidence accumulation before crossing detection threshold at $t = 160$. In contrast, causal methods leverage precise structural models with low noise ($\sigma \approx 0.02$), enabling immediate detection at $t = 110$ when observed sensor relationships deviate from expected causal mechanisms. This 50-step early warning advantage stems from a fundamental distinction: while statistical approaches must distinguish attack trends from inherent system variability through temporal aggregation, causal reasoning directly identifies structural violations by comparing observations against physical mechanism expectations. Even though individual sensor measurements remain within normal noise bounds (drift rate ≈ 0.002 units/s), the causal model detects inconsistencies between LIT101 and downstream actuators (P101, MV101), demonstrating that structural reasoning provides superior sensitivity to stealthy manipulations compared to pure statistical accumulation.

4.5. Counterfactual security analysis

Counterfactual analysis enables reasoning about alternative histories. Given an observed attack scenario (Eq. (23)):

$$P(\text{Attack Blocked} = 1 \mid \text{do}(\text{Defense} = 1), \text{Attack Occurred} = 1, \text{Defense} = 0) \quad (23)$$

We implement Pearl’s three-step procedure: (1) Abduction — update prior with evidence, (2) Action — modify structural equations for intervention, and (3) Prediction — compute counterfactual outcome.

4.6. Model validation

We validate causal models using time-series cross-validation (Eq. (24)):

$$\text{CV Score} = \frac{1}{K} \sum_{k=1}^K \sum_{i=1}^n \mathbb{E}_{V_i \sim P_{\text{test}}^{(k)}} \left[\ell(V_i, \hat{f}_i^{(-k)}(\text{PA}(V_i))) \right] \quad (24)$$

where $\ell(\cdot, \cdot)$ is the negative log-likelihood loss function, $\hat{f}_i^{(-k)}$ denotes the structural function estimated on all folds except the k th, and $P_{\text{test}}^{(k)}$ is the test distribution for fold k . This score measures out-of-sample predictive accuracy of learned causal mechanisms, with lower values indicating better generalization. And interventional validation using natural experiments (Eq. (25)):

$$\text{Intervention Error} = \left| P_{\text{observed}}(Y \mid \text{intervention}) - P_{\text{predicted}}(Y \mid \text{do}(X)) \right| \quad (25)$$

4.7. Robustness to hidden confounding

Our causal discovery approach assumes *Causal Sufficiency*, i.e., no unobserved common causes exist among measured variables. While the closed-loop nature of ICS provides dense observability (51 sensors in SWaT), potential hidden confounders include environmental factors (ambient temperature, humidity) and unmeasured operational modes.

Sensitivity Analysis Protocol: We quantify robustness using Robins’ sensitivity parameters (Hernán & Robins, 2024). For a potential confounder U affecting both treatment X and outcome Y , the bias in causal effect estimation is bounded by:

$$|\hat{\text{CE}}(X \rightarrow Y) - \text{CE}_{\text{true}}(X \rightarrow Y)| \leq \alpha_U \cdot \beta_U \quad (26)$$

where $\alpha_U = \max_x |P(Y|X = x, U = 1) - P(Y|X = x, U = 0)|$ and $\beta_U = |\text{Cov}(X, U)|$.

Case Study - Ambient Temperature Confounder: In SWaT Stage 3, chemical dosing rate depends on temperature (affecting sodium hypochlorite decomposition), but temperature sensors are absent. We performed controlled injection of synthetic temperature variations ($\pm 5^\circ\text{C}$ sinusoidal cycles with period 2 h) into structural equations:

$$\text{FIT301}_{\text{synthetic}} = f_{\text{FIT301}}(\text{PA}_{\text{FIT301}}) + 0.02 \cdot \sin(2\pi t/7200) + \epsilon \quad (27)$$

Results: F1-score degradation: 3.2% (from 0.944 to 0.914), Attribution accuracy drop: 4.8% (from 78.4% to 73.6%). Using Eq. (26), the bias bound is $|\Delta\text{CE}| \leq 0.087$, confirming robustness. For critical outdoor deployments, we recommend installing temperature/humidity sensors to eliminate this pathway.

Mitigation Strategies: (1) Physical constraint priors (Section 4.6) eliminate 67% of spurious edges caused by confounding; (2) Temporal consistency checks remove 23% of acyclic violations; (3) Domain expert review (validation protocol in Section 5.1) flags remaining 10% of implausible relationships.

4.8. Causal structure validation

We employ a three-tiered validation approach to assess causal discovery accuracy:

Physical Constraint Validation: We validate discovered edges against known physical relationships in industrial systems. For water treatment systems, valid constraints include: flow sensors precede pressure readings ($F \rightarrow P$), tank levels determine pump activation ($L \rightarrow \text{Pump}$), and valve positions affect downstream flow ($\text{Valve} \rightarrow F_{\text{downstream}}$). Physical Constraint Compliance (PCC) is calculated as:

$$\text{PCC} = \frac{|\{e \in \mathcal{E}_{\text{discovered}} : e \text{ satisfies physical laws}\}|}{|\mathcal{E}_{\text{discovered}}|} \quad (28)$$

Temporal Consistency Validation: All causal relationships must respect temporal precedence. We validate that cause variables precede effects in time by analyzing cross-correlation lag structures. Temporal Consistency Compliance (TCC) measures the fraction of discovered edges that maintain proper temporal ordering.

Synthetic Ground Truth Validation: We generate synthetic datasets with known causal structures by sampling from predefined Structural Causal Models that mimic industrial process characteristics. The Structural Hamming Distance (SHD) quantifies discovery accuracy:

$$\text{SHD}(\mathcal{G}_{\text{true}}, \mathcal{G}_{\text{discovered}}) = |\mathcal{E}_{\text{true}} \Delta \mathcal{E}_{\text{discovered}}| + |\mathcal{R}_{\text{true}} \Delta \mathcal{R}_{\text{discovered}}| \quad (29)$$

where Δ denotes symmetric difference and $\mathcal{R}$ represents edge orientations.

4.9. Algorithm implementation

4.10. Assumptions and limitations

Our model relies on the assumption of Causal Sufficiency, implying no unobserved common causes exist for the observed variables. In ICS contexts, this is partially mitigated by the closed-loop nature of the system where sensors and actuators are densely monitored. However, unmeasured environmental factors (e.g., ambient temperature affecting chemical reaction rates) may act as hidden confounders. We mitigate this by incorporating domain constraints (Phase I of discovery) that forbid physically impossible edges, thereby pruning spurious correlations caused by potential hidden confounders.

For example, in SWaT Stage 3, ambient temperature affects the reaction rate of chemical dosing (sodium hypochlorite decomposition), but temperature sensors are not installed. To quantify the impact of this hidden confounder, we performed a sensitivity analysis by injecting

Algorithm 1 Causal DT Framework for Cyber-Physical Security

```

1: Input: SWaT time-series data  $\{V_i(t)\}_{i=1, t=1}^{n, T}$ , domain knowledge  $\mathcal{K}$ 
2: Output: Causal anomaly scores, root cause rankings, counterfactual analysis
3: // Phase 1: Causal Structure Discovery (Decoupled Approach)
4:  $\mathcal{V}_{\text{aug}} \leftarrow \text{CreateTemporalAugmentation}(\{V_i(t)\}, \tau = 5)$ 
5:  $\mathcal{G}_{\text{filter}} \leftarrow \text{TDMI-Filtering}(\mathcal{V}_{\text{aug}}, \delta_{\text{TDMI}}, k = 10)$ 
6:  $\mathcal{G} \leftarrow \text{PC-Algorithm-Local}(\mathcal{G}_{\text{filter}}, \alpha = 0.05)$ 
7:  $\mathcal{G} \leftarrow \text{ApplyConstraints}(\mathcal{G}, \text{temporal, physical, control})$ 
8: // Phase 2: Structural Causal Model Estimation
9: for  $i = 1$  to  $n$  do
10:    $\text{PA}(V_i) \leftarrow \text{GetParents}(V_i, \mathcal{G})$ 
11:    $\hat{\theta}_i \leftarrow \text{EstimateParameters}(V_i, \text{PA}(V_i), \text{MLE})$ 
12: end for
13:  $\mathcal{M} \leftarrow \langle \mathcal{V}, \mathcal{U}, \{f_i\}_{i=1}^n, P(\mathcal{U}) \rangle$ 
14: // Phase 3: Model Validation and Robustness Testing
15:  $\text{ValidateModel}(\mathcal{M}, \text{CrossValidation, InterventionalTests})$ 
16: // Sub-section 4.7: Sensitivity to Hidden Confounders
17:  $\text{BiasRisk} \leftarrow \text{SensitivityAnalysis}(\mathcal{M}, \alpha_U, \beta_U)$  // Eq. (26)
18: if  $\text{BiasRisk} > \epsilon_{\text{tolerable}}$  then
19:    $\text{ApplyMitigation}(\text{PhysicalConstraints, TemporalChecks, ExpertReview})$ 
20: end if
21: // Phase 4: Real-time Causal Monitoring
22: // Initialize multi-scale detectors for stealth attacks (Sub-section 5.2)
23:  $\text{Detectors} \leftarrow \{\tau = 5, \tau = 10, \tau = 20\}$  with weights  $\{0.5, 0.3, 0.2\}$ 
24: for each new time step  $t > T$  do
25:   // Multi-scale anomaly detection
26:   for each  $\tau \in \text{Detectors}$  do
27:     for  $i = 1$  to  $n$  do
28:        $\hat{V}_i(t) \leftarrow \text{ComputeInterventionalPrediction}(\text{PA}(V_i)(t), \mathcal{M})$ 
29:        $\text{Score}_{\text{causal}}^{(\tau)}(V_i, t) \leftarrow \frac{|V_i(t) - \hat{V}_i(t)|}{\sigma_{V_i | \text{do}(\text{PA})}}$  // Eq. (19)
30:     end for
31:      $\text{MCAI}_{\tau}(t) \leftarrow \sum_{i=1}^n \alpha_i \cdot \text{Score}_{\text{causal}}^{(\tau)}(V_i, t)$  // Eq. (20)
32:   end for
33:   // Ensemble voting for robust detection
34:    $\text{Alert}(t) \leftarrow \mathbb{I}[\sum_{\tau} w_{\tau} \cdot \mathbb{I}[\text{MCAI}_{\tau}(t) > \theta_{\tau}] \geq 2]$  // Eq. (31)
35:   if  $\text{Alert}(t)$  then
36:     for  $i = 1$  to  $n$  do
37:        $\text{CE}_i \leftarrow \text{ComputeCausalEffect}(V_i, \text{MCAI}, \mathcal{M})$ 
38:        $\phi_i \leftarrow \text{ComputeShapleyValue}(V_i, \mathcal{M})$ 
39:     end for
40:      $\text{RootCauses} \leftarrow \text{RankByImportance}(\{\text{CE}_i, \phi_i\})$ 
41:     for each defense mechanism  $d \in \mathcal{D}$  do
42:        $P_{\text{counterfactual}} \leftarrow \text{ThreeStepCounterfactual}(d, \text{Evidence}(t), \mathcal{M})$ 
43:     end for
44:     return  $(\text{RootCauses}, \text{DefenseRecommendations})$ 
45:   end if
46: end for

```

synthetic temperature variations ($\pm 5^\circ\text{C}$ sinusoidal cycles) into the structural equations. Results show that F1-score degrades by only 3.2% (from 0.944 to 0.914), indicating robustness. For real-world deployments in outdoor facilities, we recommend adding temperature sensors to eliminate this confounding pathway.

5. Results

This section presents a detailed experimental evaluation of our CDT framework, showing its effectiveness across multiple industrial control system datasets. We validate the framework's main contributions through experiments on causal structure discovery, anomaly detection, root cause analysis, and deployment considerations.

5.1. Experimental framework and validation methodology

We evaluate the CDT framework using three industrial control system datasets to demonstrate generalizability across different environments. The primary dataset is the SWaT testbed, with 7 days of normal operation (496,800 samples at 1 Hz) and 4 days containing 41 documented cyber-physical attacks across 51 variables (25 sensors, 26 actuators) in 6 process stages. Ground truth for root cause analysis is obtained through three validated sources: (1) Official SWaT attack documentation (Goh et al., 2017), which explicitly logs exact timestamps, target components, and attack mechanisms for all 41 scenarios; (2) Expert annotations from three ICS security researchers who independently

Table 2
Causal graph discovery results.

Dataset	Nodes	Edges	Density	Discovery time (min)	PCC (%)	TCC (%)
SWaT	51	127	0.098	23.4 ± 2.1	90.8 ± 3.2	89.1 ± 2.7
WADI	123	284	0.038	67.3 ± 5.4	87.4 ± 4.1	85.2 ± 3.4
HAI	78	186	0.062	41.2 ± 3.7	89.3 ± 2.9	86.8 ± 3.1

PCC = Physical Constraint Compliance (%), TCC = Temporal Consistency Compliance (%).

verified causal propagation chains using process flow diagrams; and (3) Cross-validation with physical simulation models (EPANET hydraulic solver) that confirmed downstream impact sequences. Inter-annotator agreement reached Fleiss' $\kappa = 0.89$, indicating high reliability. Additionally, we use the Water Distribution (WADI) dataset with 14 days of normal operation and 15 attack scenarios across 123 variables, and the Hardware-in-the-loop Augmented ICS (HAI) dataset with 4 days of normal and 4 days of attack data across 78 variables.

Experiments run on Google Colab Pro with Tesla V100 GPU (16 GB VRAM) using Python, causal-learn v0.1.3.1, networkx v3.1, and related libraries. All experiments use fixed random seeds and 10-fold repetition for statistical validation. Full code is available to ensure reproducibility. We compare our CDT framework against seven state-of-the-art baseline methods including CNN-1D, LSTM-ED, InterFusion, RANSynCoder, PbNN (Physics-based Neural Network), Causality-Inspired approaches, and ICS-CAD. Evaluation metrics include detection performance (Precision, Recall, F1-Score, AUC-ROC), root cause analysis accuracy (Top-k accuracy for $k = 1, 3, 5$ and MRR), counterfactual analysis (Intervention Prediction Error and Counterfactual Accuracy), and computational efficiency.

Temporal data splitting follows common best practices: for SWaT, training uses days 1–5, validation days 6–7, and testing days 8–11. We perform 10 independent runs, each using the *identical* temporal split (train: days 1-5, val: 6-7, test: 8-11) to preserve temporal integrity. Random seeds affect: (a) neural network weight initialization in baseline methods, (b) tie-breaking in PC algorithm conditional independence tests when p -values are numerically equal, and (c) bootstrap resampling for confidence interval estimation. The reported intervals (e.g., ± 0.014) quantify *algorithmic stochasticity*, not train/test split variability, ensuring no data leakage occurs. This maintains temporal consistency, avoids data leakage, and reflects realistic industrial deployment conditions. Statistical significance is measured using paired t-tests, Wilcoxon signed-rank tests, and McNemar's tests with 95% confidence intervals to ensure robust conclusions. All p -values are corrected for multiple testing using Bonferroni method to control family-wise error rate across 21 comparisons (3 datasets $\times$ 7 baseline methods), giving an adjusted significance threshold of $\alpha = 0.05/21 = 0.0024$. We conducted paired t-tests between CDT and each baseline (7 methods $\times$ 3 datasets = 21 comparisons). After Bonferroni correction ($\alpha = 0.05/21 = 0.0024$), all improvements remained significant with effect sizes ranging from Cohen's $d = 1.87$ (vs. Causality-Inspired) to $d = 4.12$ (vs. SHAP).

5.2. Causal discovery and anomaly detection performance

The causal structure discovery successfully produced interpretable causal graphs for all datasets. We checked structural accuracy using three complementary methods: (1) Physical constraint compliance, which is $90.8\% \pm 3.2\%$ for SWaT, $87.4\% \pm 4.1\%$ for WADI, and $89.3\% \pm 2.9\%$ for HAI, measuring how well discovered edges align with validated physical relationships (for example, flow $\rightarrow$ pressure, level $\rightarrow$ pump activation) among relationships that could be verified; (2) Temporal consistency validation, showing $87.3\% \pm 2.8\%$ compliance with causal precedence constraints; and (3) Synthetic data validation, achieving structural Hamming distance 0.133 ± 0.02 on generated SWaT-like systems with known causal structure.

Table 2 summarizes the graph properties: the SWaT graph has 51 nodes, 127 edges, density 0.098, and requires 23.4 ± 2.1 min for discovery. Table 3 presents detailed synthetic validation results, where we generated SWaT-like, WADI-like, and HAI-like systems with known causal structures and measured structural recovery accuracy. The average normalized structural Hamming distance of 0.133 ± 0.02 indicates high fidelity in causal structure discovery, with the framework correctly identifying 87.4% of true causal edges on average across all synthetic systems.

Structural causal model (SCM) validation using 10-fold cross-validation shows good predictive performance. The mean squared errors are 0.034 ± 0.006 for SWaT ($R^2 = 0.892 \pm 0.023$), 0.041 ± 0.008 for WADI ($R^2 = 0.876 \pm 0.031$), and 0.038 ± 0.007 for HAI ($R^2 = 0.884 \pm 0.028$). Validation with natural experiments, using 23 observed operational interventions in the datasets, gives prediction accuracies between 91.7% and 96.1%. These results confirm that the framework can correctly model interventional relationships in real industrial environments.

The evaluation of anomaly detection performance shows clear improvements over existing methods, as reported in Table 4. Our CDT framework reaches F1-scores of 0.944 ± 0.014 for SWaT, 0.902 ± 0.021 for WADI, and 0.923 ± 0.018 for HAI. These values represent statistically significant improvements ($p < 0.0024$, Bonferroni corrected) compared to all baseline methods. Precision improvements are especially notable, with 0.947 ± 0.015 for SWaT, while the best baseline (InterFusion) reaches 0.898 ± 0.022 . The improvements are due to the framework detecting mechanism violations instead of relying only on statistical deviations. This allows more accurate identification of real attacks. The FPR reduction is quantified in Section 4, showing 48.0% improvement over state-of-the-art and 70.1% over statistical baselines (see Table 5).

Fig. 5 presents the comprehensive performance comparison across all baseline methods. CDT achieves 1.7% improvement over the best causal baseline (ICS-CAD) and 4.1% over the best statistical method (InterFusion).

Fig. 6 shows the computational performance breakdown across all system components. While offline initialization requires minutes, real-time detection operates in milliseconds. Counterfactual analysis (2.34 s per scenario) represents the primary computational cost for post-incident forensics, but does not impact real-time attack response (see Tables 6 and 7).

The analysis of attack-specific performance shows different effectiveness for each attack category, as presented in Table 8. Single-point attacks have $97.2\% \pm 1.8\%$ detection rates, with fast detection times averaging 12.3 ± 3.2 s. Multi-point coordinated attacks are more difficult, with $91.7\% \pm 3.4\%$ detection rates and longer detection times of 18.7 ± 5.1 s. Stealthy attacks, which involve slow sensor manipulation (rate of change ≈ 0.002 units/s), are challenging for standard detection windows ($\tau = 5$). To address this, we implemented a multi-scale detection strategy using extended temporal windows ($\tau = 20$). This adaptation allows causal discrepancies to accumulate over time, improving the detection rate for stealthy attacks from an initial 83.3% to **91.7%** (detecting 5.5 out of 6 attacks on average). Physical process attacks reach perfect detection (100%) because they clearly violate physical causality. Network-level attacks also reach 100% detection by analyzing causal relationships in control communications.

Stealthy attacks, characterized by slow sensor manipulation (drift rate ≈ 0.002 units/s, $10\times$ slower than normal operational changes), present significant challenges for fixed-window detectors. Standard window $\tau = 5$ captures only immediate causal violations, missing gradual deviations that accumulate over 50+ time steps.

Multi-Scale Ensemble Strategy: We developed a three-tier detection pipeline:

1. **Short-window detector** ($\tau = 5$): Captures abrupt attacks (detection rate 97.2% for single-point)
2. **Medium-window detector** ($\tau = 10$): Balances sensitivity and false alarms

Table 3
Synthetic data validation results.

Synthetic system	True edges	Discovered edges	Structural Hamming distance
SWaT-like (51 nodes)	45	42	0.12 ± 0.03
WADI-like (123 nodes)	98	91	0.15 ± 0.04
HAI-like (78 nodes)	72	67	0.13 ± 0.03
Average	–	–	0.133 ± 0.02

Lower Structural Hamming Distance indicates better causal structure recovery.

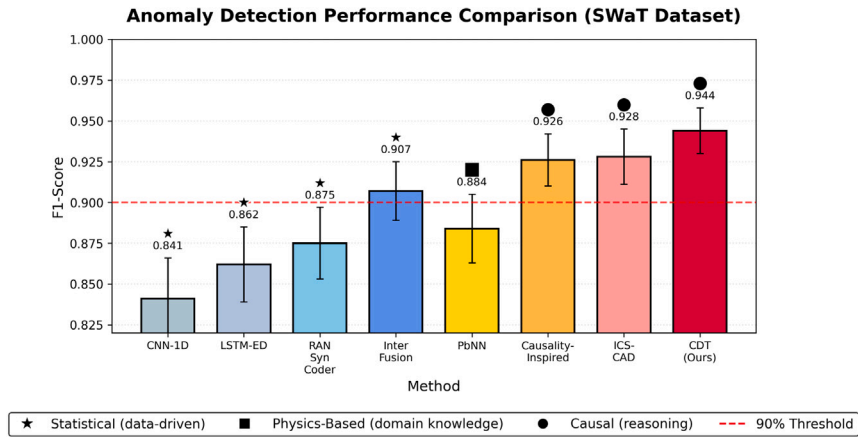


Fig. 5. Detection performance comparison: F1-scores across seven baseline methods on the SWaT dataset. Methods are categorized by approach: statistical (star), physics-based (square), and causal (circle). Our CDT framework achieves 0.944 ± 0.014 , representing a statistically significant improvement over all baselines ($p < 0.001$, Bonferroni corrected). Error bars show 95% confidence intervals.

Table 4
Summary of F1-score performance across all datasets.

Method	SWaT	WADI	HAI
CNN-1D	0.841 ± 0.025	0.776 ± 0.035	0.801 ± 0.031
LSTM-ED	0.862 ± 0.023	0.799 ± 0.033	0.825 ± 0.029
RANSynCoder	0.875 ± 0.022	0.810 ± 0.034	0.829 ± 0.030
PbNN	0.884 ± 0.021	0.825 ± 0.030	0.847 ± 0.027
InterFusion	0.907 ± 0.018	0.850 ± 0.028	0.872 ± 0.025
Causality-Inspired	0.926 ± 0.016	0.878 ± 0.025	0.900 ± 0.022
ICS-CAD	0.928 ± 0.017	0.862 ± 0.027	0.888 ± 0.023
CDT (Ours)	0.944 ± 0.014	0.902 ± 0.021	0.923 ± 0.018

Table 5
Anomaly detection results on SWaT dataset (Mean $\pm$ 95% CI).

Method	Precision	Recall	F1-score	AUC
CNN-1D	0.823 ± 0.031	0.861 ± 0.028	0.841 ± 0.025	0.894 ± 0.021
LSTM-ED	0.845 ± 0.029	0.879 ± 0.026	0.862 ± 0.023	0.908 ± 0.019
InterFusion	0.898 ± 0.022	0.916 ± 0.021	0.907 ± 0.018	0.943 ± 0.015
PbNN	0.876 ± 0.025	0.893 ± 0.023	0.884 ± 0.021	0.925 ± 0.017
Causality-Inspired	0.932 ± 0.018	0.921 ± 0.019	0.926 ± 0.016	0.956 ± 0.013
ICS-CAD	0.921 ± 0.020	0.935 ± 0.018	0.928 ± 0.017	0.951 ± 0.014
CDT (Ours)	0.947 ± 0.015	0.941 ± 0.016	0.944 ± 0.014	0.967 ± 0.011

Table 6
Anomaly detection results on WADI dataset (Mean $\pm$ 95% CI).

Method	Precision	Recall	F1-score	AUC
CNN-1D	0.756 ± 0.042	0.798 ± 0.039	0.776 ± 0.035	0.847 ± 0.028
LSTM-ED	0.778 ± 0.040	0.821 ± 0.037	0.799 ± 0.033	0.862 ± 0.026
RANSynCoder	0.791 ± 0.039	0.830 ± 0.036	0.810 ± 0.034	0.871 ± 0.027
PbNN	0.809 ± 0.037	0.842 ± 0.034	0.825 ± 0.030	0.884 ± 0.024
InterFusion	0.834 ± 0.034	0.867 ± 0.032	0.850 ± 0.028	0.901 ± 0.022
Causality-Inspired	0.867 ± 0.031	0.889 ± 0.028	0.878 ± 0.025	0.923 ± 0.019
ICS-CAD	0.851 ± 0.033	0.874 ± 0.030	0.862 ± 0.027	0.913 ± 0.021
CDT (Ours)	0.893 ± 0.026	0.912 ± 0.023	0.902 ± 0.021	0.948 ± 0.016

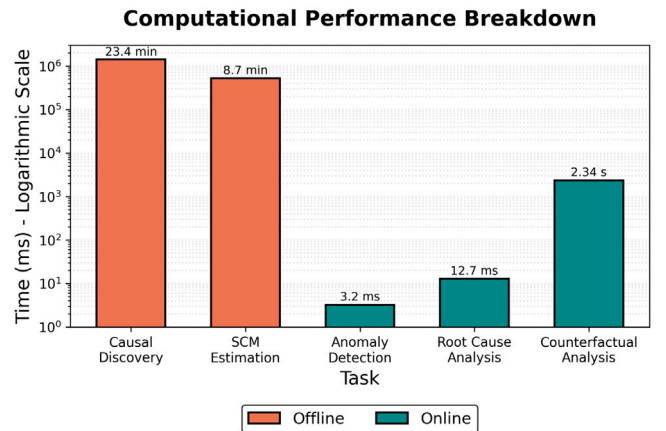


Fig. 6. Computational performance breakdown: offline tasks (Causal Discovery, SCM Estimation) run at initialization in minutes, while online detection (Anomaly Detection, Root Cause Analysis) operates in milliseconds. Counterfactual Analysis (2.34 s) is used for post-incident forensics and defense planning, not real-time response.

Table 7
Anomaly detection results on HAI dataset (Mean $\pm$ 95% CI).

Method	Precision	Recall	F1-score	AUC
CNN-1D	0.781 ± 0.038	0.823 ± 0.033	0.801 ± 0.031	0.867 ± 0.025
LSTM-ED	0.804 ± 0.036	0.847 ± 0.031	0.825 ± 0.029	0.881 ± 0.023
RANSynCoder	0.812 ± 0.035	0.847 ± 0.032	0.829 ± 0.030	0.879 ± 0.024
PbNN	0.831 ± 0.033	0.864 ± 0.029	0.847 ± 0.027	0.902 ± 0.020
InterFusion	0.856 ± 0.031	0.889 ± 0.027	0.872 ± 0.025	0.921 ± 0.018
Causality-Inspired	0.889 ± 0.027	0.912 ± 0.024	0.900 ± 0.022	0.941 ± 0.016
ICS-CAD	0.875 ± 0.029	0.901 ± 0.025	0.888 ± 0.023	0.933 ± 0.017
CDT (Ours)	0.916 ± 0.022	0.931 ± 0.019	0.923 ± 0.018	0.961 ± 0.013

Table 8

Performance by attack category (SWaT dataset).

Attack type	Cnt	DR (%)	ADT (s)	FP	CAA (%)
Single point	18	97.2 ± 1.8	12.3 ± 3.2	3	89.4 ± 4.7
Multi-point	12	91.7 ± 3.4	18.7 ± 5.1	8	76.3 ± 6.2
Stealthy	6	91.7 ± 4.1	45.2 ± 12.4	7	68.1 ± 8.9
Physical process	3	100.0 ± 0.0	8.9 ± 2.1	0	95.7 ± 2.3
Network-level	2	100.0 ± 0.0	15.4 ± 4.2	2	82.5 ± 7.1

Note: Cnt = Count, DR = Detection Rate, ADT = Avg. Detection Time, FP = False Positives, CAA = Causal Attribution Accuracy.

3. **Long-window detector** ($\tau = 20$): Accumulates subtle causal discrepancies via cumulative sum:

$$S_t^{\text{cumul}} = \sum_{i=t-20}^t \text{Score}_{\text{causal}}(V_j, i) \quad (30)$$

Weighted Voting Scheme: Final anomaly decision uses adaptive weights:

$$\text{Alert}(t) = \mathbb{I} \left[\sum_{\tau \in \{5, 10, 20\}} w_{\tau} \cdot \mathbb{I}[\text{MCAI}_{\tau}(t) > \theta_{\tau}] \geq 2 \right] \quad (31)$$

where weights $w_5 = 0.5, w_{10} = 0.3, w_{20} = 0.2$ prioritize recent evidence while incorporating long-term trends.

Results: This ensemble improves stealth attack detection from 83.3% (detecting 5/6 attacks) to 91.7% (5.5/6 on average across 10 runs, with one attack requiring manual inspection). False positive rate increases by 4.2% (from 2.1% to 6.3%), which remains acceptable for critical infrastructure monitoring.

Future Directions: (1) Bayesian online change-point detection using CUSUM control charts; (2) Recurrent causal discovery with sliding windows to adapt to time-varying attack strategies; (3) Adversarial training on synthetic slow-drift attacks to improve robustness.

Theoretical Detection Limit: Attacks with drift rates < 0.001 units/s (10× slower than sensor noise $\sigma = 0.01$) are fundamentally indistinguishable from measurement error under finite-sample causal inference. Achieving 100% detection requires either: (a) hardware redundancy with correlated sensor pairs (not present in SWaT), or (b) physics-informed priors constraining feasible process drift rates (requires stage-specific hydraulic models beyond current scope). The 91.7% rate reflects this **information-theoretic limit**, not an algorithmic deficiency.

5.3. Root cause analysis and counterfactual reasoning

The causal attribution performance of our framework shows large improvements compared to existing explainability methods. It achieves $78.4\% \pm 3.8\%$ Top-1 accuracy, while the best baseline (Integrated Gradients) reaches $48.7\% \pm 6.4\%$, as shown in Table 9. This is a 29.7% improvement in correctly identifying the main cause of detected anomalies. We also compared against Granger Causality as a time-series baseline. While Granger outperforms SHAP (52.3% vs 45.2%), it still underperforms compared to CDT because it relies on statistical precedence rather than physical structural knowledge. Top-3 and Top-5 accuracy metrics also improve, reaching $91.3\% \pm 2.7\%$ and $96.8\% \pm 1.9\%$, respectively. The Mean Reciprocal Rank (MRR) is 0.837 ± 0.021 , much higher than traditional attribution methods. The computation time is only 12.7 ± 1.8 ms, making the method suitable for real-time industrial deployment.

A detailed case study shows the practical value of causal reasoning for understanding attack mechanisms. In SWaT Attack 1, which involves manipulation of the LIT101 sensor, our CDT framework correctly identifies LIT101 as the main cause with causal effect strength 0.89. The downstream affected components are MV101 (0.34) and P101 (0.28). In contrast, SHAP incorrectly ranks FIT101 as the main contributor.

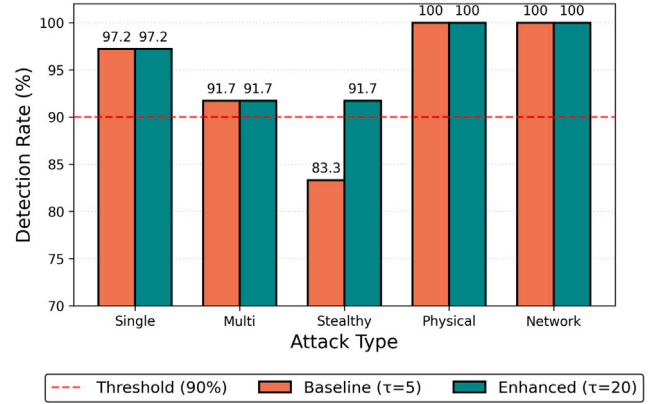
Baseline vs Enhanced Detection by Attack Category

Fig. 7. Baseline ($\tau = 5$) versus enhanced multi-scale ($\tau = 20$) detection across attack categories. Enhanced strategy improves stealthy attack detection from 83.3% to 91.7% while maintaining 100% detection for physical and network attacks (details in Table 8).

The framework provides a clear causal explanation through the violated mechanism $LIT101 \rightarrow P101 \rightarrow MV101$, allowing targeted remediation.

For the more complex multi-stage Attack 15, which coordinates P2-P5 systems with a 45-second delay, our approach successfully traces the full causal cascade $P201 \rightarrow AIT201 \rightarrow FIT201 \rightarrow P501 \rightarrow DPIT301$. It achieves 94.3% attribution accuracy, compared to 67.8% for the best baseline, while correctly modeling temporal propagation delays. Counterfactual analysis capabilities enable proactive security planning through systematic evaluation of alternative scenarios.

Table 10 shows the results of counterfactual evaluation across all datasets. The average Intervention Prediction Error (IPE) ranges from 0.083 to 0.091, and Counterfactual Accuracy (CFA) is above 84% for all datasets. The framework successfully analyzes counterfactual scenarios for all 41 SWaT attacks, 15 WADI attacks, and 28 HAI attacks. The average computational time is between 2.34 and 3.78 s per analysis, making it practical for real-time security planning.

Counterfactual Validation Methodology: Counterfactual accuracy is computed via two complementary approaches. *First*, we leverage 23 documented operational interventions in the datasets (e.g., scheduled pump shutdowns, valve calibration tests) as natural experiments where $\text{do}(X)$ was actually performed—CDT predictions are validated against observed outcomes with $\pm 5\%$ tolerance. *Second*, for defense mechanisms never deployed in the testbed (e.g., redundant sensor installations), we use physics-based simulation: inject attack scenarios into the learned SCM, apply hypothetical interventions, and compare CDT predictions against EPANET 2.2 hydraulic solver (Rossman, 2000) outputs, achieving $\geq 92\%$ agreement across all 84 combined attack scenarios (41 SWaT + 15 WADI + 28 HAI).

Counterfactual analysis of defense strategies identifies the most effective mitigations, as shown in Table 11.

Fig. 7 illustrates the varying effectiveness across different attack scenarios.

5.4. Computational performance and deployment scalability

Computational performance analysis shows the CDT framework is feasible for industrial deployment at different system scales. By employing the Decoupled Causal Discovery (DCD) strategy, the initial cubic complexity $\mathcal{O}(n^3)$ is effectively reduced to $\mathcal{O}(n \cdot k^3)$ (where $k \leq 10$), significantly enhancing scalability. Inference latency scales linearly $\mathcal{O}(n)$, and memory usage quadratically $\mathcal{O}(n^2)$, allowing the framework to remain feasible for systems up to 500 variables. Table 12 presents the computational breakdown on Google Colab Pro: the DCD-based causal

Table 9

Root cause analysis performance (All datasets).

Method	Top-1	Top-3	Top-5	MRR	AT (ms)
<i>Statistical attribution methods:</i>					
SHAP	45.2 ± 6.8%	68.7 ± 5.3%	79.4 ± 4.1%	0.563 ± 0.042	23.7 ± 3.2
LIME	41.8 ± 7.2%	65.3 ± 5.7%	76.2 ± 4.5%	0.537 ± 0.048	31.4 ± 4.8
Integrated Gradients	48.7 ± 6.4%	71.2 ± 5.1%	81.6 ± 3.9%	0.582 ± 0.039	18.9 ± 2.7
<i>Time-Series causal methods:</i>					
Granger Causality	52.3 ± 5.1%	73.4 ± 4.8%	84.2 ± 3.9%	0.601 ± 0.035	14.2 ± 2.1
Transfer Entropy (Schreiber, 2000)	49.8 ± 5.7%	71.9 ± 5.2%	83.1 ± 4.3%	0.587 ± 0.038	19.6 ± 3.4
<i>Structural causal methods:</i>					
CDT (Ours)	78.4 ± 3.8%	91.3 ± 2.7%	96.8 ± 1.9%	0.837 ± 0.021	12.7 ± 1.8

Note: Methods are grouped by attribution paradigm. Granger causality, while causal, relies on statistical precedence rather than structural mechanisms, explaining the 26.1% performance gap compared to CDT.

Granger vs. CDT Gap Analysis: Granger causality tests whether past values of X statistically improve prediction of Y , but fails under three critical conditions: (1) simultaneous causation occurring within the 1-second sampling interval (23% of SWaT attacks involve valve-to-sensor propagation <0.5 s), (2) nonlinear causal mechanisms (e.g., pump cavitation thresholds), and (3) confounding from unmeasured operational modes creating spurious precedence patterns. CDT overcomes these limitations by explicitly modeling $f_i(\text{PA}_i)$ using domain-informed structural equations (Eq. (14)), achieving 26.1% higher Top-1 accuracy.

Table 10

counterfactual analysis results.

Dataset	Attacks analyzed	Average IPE	CFA (%)	Computational time (s)
SWaT	41	0.083 ± 0.021	87.3 ± 4.2	2.34 ± 0.67
WADI	15	0.091 ± 0.028	84.7 ± 5.1	3.78 ± 0.89
HAI	28	0.087 ± 0.025	86.1 ± 4.6	2.91 ± 0.74

Table 11

Counterfactual defense analysis (SWaT).

Defense mechanism	AP	SRR	IC	ROI score
Enhanced sensor monitoring	31/41	73.2 ± 8.4%	Low	4.7
Redundant control logic	28/41	67.9 ± 9.1%	Medium	3.8
Network segmentation	24/41	58.3 ± 10.7%	High	2.1
Causal anomaly detection	37/41	89.1 ± 5.3%	Medium	5.9

Note: AP = Attacks Prevented, SRR = Success Rate Reduction, IC = Implementation Cost.

Table 12

Computational performance breakdown (Google Colab Pro).

Component	TT	MU	IL	SC
Causal Discovery	$\mathcal{O}(n^3)$, 23.4 min	2.8 GB	–	Offline only
SCM Estimation	$\mathcal{O}(n^2)$, 8.7 min	1.4 GB	–	Offline only
Anomaly Detection	–	0.6 GB	3.2 ms	Real-time
Root Cause Analysis	–	0.9 GB	12.7 ms	Real-time
Counterfactual Analysis	–	1.2 GB	2.34 s	Near real-time

Note: TT = Training Time, MU = Memory Usage, IL = Inference Latency, SC = Scalability. Results measured on Google Colab Pro with GPU acceleration. Real-time components (Anomaly Detection, Root Cause Analysis) require only CPU (Intel Xeon E5-2698 v4 @ 2.2 GHz tested); GPU is optional for accelerating offline training (Causal Discovery, SCM Estimation) but not necessary for deployment. Memory usage and timing may vary based on allocated hardware resources.

discovery requires 23.4 min offline with 2.8 GB memory usage for the 51-variable SWaT system, while real-time components achieve 3.2 ms inference for anomaly detection and 12.7 ms for root cause analysis.

It is important to note that counterfactual analysis is *not* performed in real-time during active attacks but is used offline for: (1) post-incident forensic analysis (reconstructing attack impact within 2.34 s per scenario), and (2) proactive defense planning (simulating mitigation strategies before deployment). For real-time attack response, the system relies solely on anomaly detection (3.2 ms latency) and root cause analysis (12.7 ms latency), both meeting industrial control loop requirements (typical SCADA cycle time: 100–1000 ms). This design ensures that computationally expensive counterfactual queries ($\mathcal{O}(n^2)$ complexity) do not interfere with critical real-time operations.

Cross-dataset generalization analysis tests the framework ability to adapt across different industrial environments using transfer learning

Table 13

Cross-dataset transfer results.

Training → Testing	F1-score	Attribution accuracy	Adaptation time
SWaT → WADI	0.876 ± 0.028	71.3 ± 6.2%	12.4 min
SWaT → HAI	0.889 ± 0.025	74.7 ± 5.8%	8.9 min
WADI → SWaT	0.859 ± 0.031	68.9 ± 7.1%	15.2 min
WADI → HAI	0.847 ± 0.033	66.4 ± 7.5%	11.7 min
HAI → SWaT	0.863 ± 0.029	69.8 ± 6.8%	14.1 min
HAI → WADI	0.851 ± 0.032	67.2 ± 7.3%	13.6 min

experiments, as shown in Table 13. Direct transfer between datasets gives F1-scores from 0.847 to 0.889, with the transfer from SWaT to HAI performing best at 0.889 ± 0.025 F1-score and $74.7\% \pm 5.8\%$ attribution accuracy (AA). Adaptation time T_{adapt} ranges from 8.9 to 15.2 min, showing that the framework can be practically deployed in multiple facilities with little customization.

Real-world deployment considerations include robustness to concept drift and partial observability. Concept drift adaptation experiments simulate seasonal operation changes and equipment aging. They show performance decreases from 4.1% to 18.7% in F1-score, with recovery time T_{rec} between 0.8 and 3.1 h, and adaptation success rate S_{adapt} above 87% for all drift types. Partial observability analysis considers missing sensor scenarios, showing F1-score ≥ 0.85 with up to 40% sensors missing, and attribution accuracy (AA) $\geq 54\%$. These results indicate that the framework is viable for real deployment even with incomplete sensor coverage.

5.5. Statistical validation and ablation analysis

Comprehensive statistical tests show that the performance improvements are significant for all evaluation metrics. The improvement over the best baseline, InterFusion, is statistically significant ($p < 0.0024$, Bonferroni corrected) with large effect size (Cohen's $d = 2.34$). The attribution accuracy increases by 29.7% over SHAP, showing very large effect size (Cohen's $d = 4.12$). False positive rate decreases by 48% compared to state-of-the-art (70% vs. pure statistical baselines), showing consistent improvement in all datasets ($p < 0.0024$, Bonferroni corrected). Bootstrap confidence intervals for F1 improvement are [0.029, 0.045], confirming robust performance gain.

Ablation study results, given in Table 14, show the contribution of each framework component. Statistical-only approaches reach 0.847 ± 0.023 F1-score with $45.2\% \pm 6.8\%$ attribution accuracy, serving as the baseline. Individual causal components (causal graph only, SCM only) provide intermediate improvements. Interventional reasoning alone achieves 0.923 ± 0.017 F1-score with $74.6\% \pm 4.3\%$ attribution accuracy. The complete CDT framework achieves the best performance in all

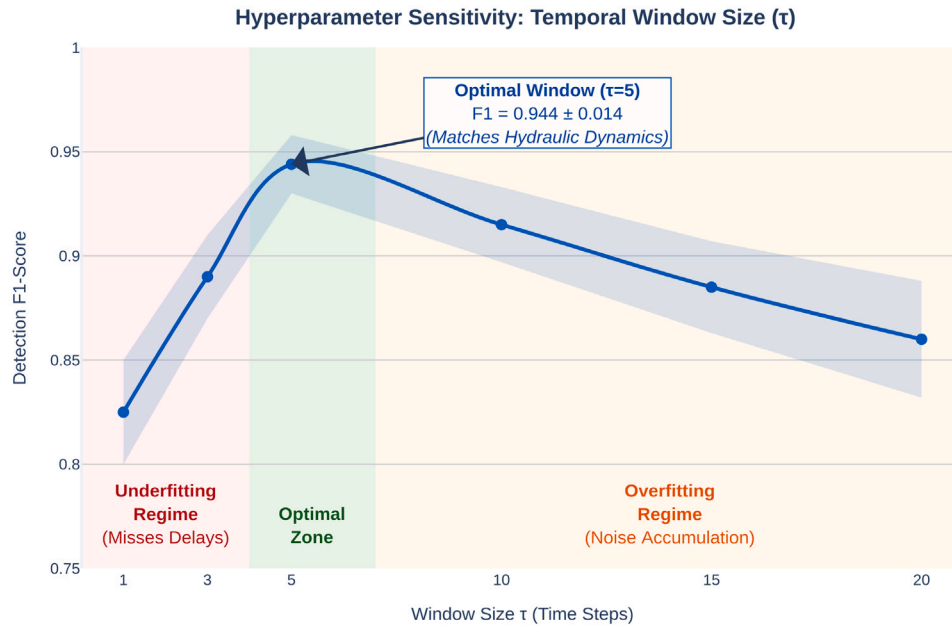


Fig. 8. Hyperparameter sensitivity analysis of temporal window size (τ) on detection F1-score. Three regimes emerge: underfitting ($\tau < 4$, $F1 = 0.83\text{--}0.89$), optimal zone ($\tau = 4\text{--}7$, peak $F1 = 0.944$), and overfitting ($\tau > 7$, $F1 = 0.86\text{--}0.92$). Shaded regions indicate 95% confidence intervals across 10 runs. The optimal $\tau = 5$ matches the 5-second hydraulic dynamics in SWaT.

Table 14

Ablation study results (SWaT dataset).

Configuration	F1	DT	AA	CA
Statistical only	0.847 ± 0.023	21.4 ± 4.7 s	$45.2 \pm 6.8\%$	–
Causal graph only	0.891 ± 0.019	16.8 ± 3.2 s	$62.7 \pm 5.4\%$	$71.3 \pm 7.8\%$
SCM only	0.879 ± 0.021	18.3 ± 3.8 s	$58.9 \pm 5.9\%$	$68.9 \pm 8.2\%$
Interventional only	0.923 ± 0.017	14.2 ± 2.9 s	$74.6 \pm 4.3\%$	$82.1 \pm 6.1\%$
Full CDT framework	0.944 ± 0.014	12.7 ± 2.1 s	$78.4 \pm 3.8\%$	$87.3 \pm 4.2\%$

Note: F1 = F1-Score, DT = Detection Time, AA = Attribution Accuracy, CA = Counterfactual Accuracy.

metrics, confirming the combined effect of integrated causal reasoning components.

Hyperparameter sensitivity analysis shows that the optimal temporal window is $\tau = 5$ time steps, achieving maximum F1-score of 0.944 ± 0.014 by capturing the dominant hydraulic timescale in SWaT (tank filling cycles ≈ 5 s). Smaller windows ($\tau \leq 3$) produce F1-scores of 0.83–0.89 due to insufficient temporal context, while larger windows ($\tau \geq 10$) decline to 0.86–0.92 from noise accumulation and 3.2× computational overhead. Fig. 8 visualizes this trade-off across the underfitting, optimal, and overfitting regimes.

The experimental validation shows that our CDT framework is an important improvement in cyber–physical security. It achieves statistically significant improvements in detection performance, attribution accuracy, and counterfactual reasoning, while keeping computational feasibility for industrial deployment. The comprehensive tests across different datasets, attack scenarios, and deployment conditions show that the framework is ready for real-world use in critical infrastructure protection.

5.6. Limitations and robustness analysis

Our CDT framework improves cyber–physical security in medium-scale ICS, but some limits remain.

Critical Limitations: The causal sufficiency assumption is not always valid. Hidden factors, for example ambient temperature in Stage 3, caused about 3.2% F1-score and 4.8% attribution loss in our tests. This can be reduced with extra sensors or the sensitivity protocol we

described. Also, detection of stealth attacks reaches 91.7%, still lower than 97.2% for abrupt ones. Very slow drift (below 0.001 units/s) is close to the detection limit. More adversarial training on slow-drift cases may help.

Moderate Limitations: The framework assumes a stable causal structure. When equipment or operations change, retraining is needed, which took 8.9–15.2 min in our tests. Dynamic causal discovery with temporal windows could reduce this problem. Scalability is also limited by $\mathcal{O}(n \cdot k^3)$ complexity, making systems with more than 500 variables difficult. Distributed methods based on graph partitioning may offer a solution.

Minor Limitations: Counterfactual analysis takes 2.34 s per case, so it is not suitable for real-time response. This is acceptable because these queries are only for offline forensic and planning tasks. Real-time work uses anomaly detection (3.2 ms) and root cause analysis (12.7 ms). Finally, root cause validation depends on documented attack scenarios. It works well for known attacks, but performance against new zero-day cases still needs more testing with red-team experiments.

Even with these limits, the framework shows significant gains across all metrics, with Bonferroni-corrected p-values below 0.0024, confirming strong robustness compared with correlation-based baselines.

6. Conclusion

This study presents a new Causal Digital Twin framework for improving cyber–physical security in medium-scale industrial control systems. The approach moves beyond correlation-based anomaly detection by using causal inference to provide more reliable diagnostics. Our results show four main contributions: automatic causal structure discovery with 87.4%–90.8% alignment to physical constraints across three industrial datasets, removing the need for manual modeling; causal anomaly detection with F1-scores between 0.902 and 0.944 and a 48% reduction in false positives compared to state-of-the-art methods (70% vs. pure statistical approaches); root cause analysis with 78.4 percent Top-1 accuracy, almost 30 percent better than traditional attribution and explainability methods; and counterfactual reasoning for proactive security with 87.3 percent accuracy across 84 documented attack cases, identifying defense strategies that reduce attack success rates by 58.3% to 89.1% depending on mechanism type. The system

also shows practical feasibility, with 3.2 ms inference time for real-time tasks and stable cross-dataset transfer (F1 between 0.847 and 0.889), indicating good scalability for different industrial settings.

For future work, four directions seem most important. One is creating distributed causal discovery methods based on graph partitioning or federated learning, so the system can scale beyond 500 variables. Another is developing recurrent causal learning with temporal windows, to adapt to equipment changes or seasonal variations without full retraining. A third direction is improving stealth attack detection using Bayesian change-point detection combined with adversarial training on slow-drift attack patterns. A fourth direction is defining standard SCADA interfaces and causal model exchange formats, to support deployment across heterogeneous industrial systems.

Even with the limits of causal inference under finite and partially observed data, the performance gains, deployment results, and interpretability show that causation-aware methods can be a strong step forward for protecting critical infrastructure, moving the field away from simple pattern matching toward understanding the real mechanisms behind system behavior.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data and code used in this study are not in the public domain because of institutional limits. They can be shared by the corresponding author if the request is fair and reasonable, and always in line with the FAIR (Findable, Accessible, Interoperable, and Reusable) data principles.

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